

INTRODUCTION TO NMAP: PORT SCANNING SIMPLIFIED



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Introduction to Port Scanning

Port scanning is a technique used to identify open ports and services available on a networked system. It is often a preliminary step in vulnerability assessments and penetration testing.

Purposes:

- Discover available services
- Identify security risks
- Create network inventory
- Troubleshoot firewall configurations

Overview of Nmap

Nmap (Network Mapper) is a powerful open-source tool for network exploration and security auditing.

Key Features:

- Host discovery
- Port scanning
- OS and service fingerprinting
- Version detection
- Scriptable scanning with NSE

Official Website: <https://nmap.org>

3] Installation and Setup

Windows:

- Download the Nmap installer from the [official website](https://nmap.org)
- Includes Zenmap GUI

Linux:

```
sudo apt install nmap # Debian/Ubuntu
```

```
sudo yum install nmap # RedHat/CentOS
```



macOS:

```
brew install nmap    # via Homebrew
```

Nmap Basics and Syntax

Syntax Format:

```
nmap [scan types] [options] [targets]
```

Common Options:

Option	Description
-sS	SYN (Stealth) Scan
-sT	TCP Connect Scan
-p	Specify ports
-A	Enable OS detection and version detection
-T0 to -T5	Set scan speed
-Pn	Skip host discovery (treat all hosts as up)

Understanding Port States

Port State	Meaning
Open	Application is accepting connections
Closed	No application is accepting connections
Filtered	Firewall blocking access
Unfiltered	Nmap can't determine state
Open	Filtered
Closed	Filtered

Nmap Scan Types

Scan Type	Command	Description
SYN Scan	-sS	Stealthy, fast, does not complete handshake
TCP Connect	-sT	Full connection (detectable)
UDP Scan	-sU	Checks for open UDP ports
Ping Scan	-sn	Identifies live hosts
FIN Scan	-sF	Sends FIN packets to bypass filters
Xmas Scan	-sX	FIN, URG, PSH flags – stealthy
Null Scan	-sN	No TCP flags set

Advanced Scanning Techniques

- OS Detection: `nmap -O target_ip`
- Service Version Detection: `nmap -sV target_ip`
- Aggressive Scan: `nmap -A target_ip`
- Scan Specific Port Range: `nmap -p 1-1000 target_ip`
- Timing Control: `nmap -T4 target_ip` (T0 to T5)

Interpreting Nmap Output

Typical scan result includes:

- Status of host (Up/Down)
- Open/filtered/closed ports
- Detected services and versions
- OS fingerprint

Traceroute (if enabled)



Firewall Evasion Techniques

Technique	Command
Packet Fragmentation	-f
Decoy Scanning	--decoy <IP>
MAC Address Spoofing	--spoof-mac 0
Randomized Scan Order	--randomize-hosts
Custom Data Length	--data-length <number>

Nmap Scripting Engine (NSE)

NSE allows advanced scanning via community and custom scripts.

Common NSE Categories:

- auth – authentication checks
- default – safe and useful
- exploit – known vulnerabilities
- vuln – potential vulnerabilities
- malware – indicators of compromise

Usage Example:

```
nmap --script=http-enum target_ip
```

```
nmap --script=vuln -p80,443 target_ip
```

Real-World Use Cases

IT Professionals – Network monitoring, security audit

Penetration Testers – Reconnaissance and vulnerability checks

Companies – Periodic infrastructure security scans

SOC Teams – Detecting rogue services or hosts

Practical Lab Exercises

Lab : Basic Scan

```
nmap 192.168.1.1
```

Lab : Scan Specific Ports

```
nmap -p 22,80,443 192.168.1.1
```

Lab 3: Service Version Detection

```
nmap -sV 192.168.1.1
```

Lab : OS Detection + Traceroute

```
nmap -O --traceroute 192.168.1.1
```

Lab : Vulnerability Scanning with NSE

```
nmap --script=vuln 192.168.1.1
```

3 Tables and Charts

Scan Type Stealth Comparison

Scan Type	Stealthy?
SYN (-sS)	Yes
TCP (-sT)	No
FIN (-sF)	Yes
Xmas (-sX)	Yes
Null (-sN)	Yes

Timing Options

Option	Speed	Use Case Avoid IDS
-T0 -T1	Very Slow	detection Low-
-T2 -T3	Slow	impact scans Non-
-T4 -T5	Polite	intrusive Default
	Normal	Aggressive scans
	Fast	Risky, detectable
	Very Fast	

Best Practices

- ✓ Always have legal authorization before scanning
- ✓ Start with ping scan to detect live hosts
- ✓ Use -v or -d for verbose/debug output
- ✓ Combine with Wireshark or Metasploit
- ✓ Document findings for reporting

Common Errors and Troubleshooting

Problem Host shows	Solution
as down Scan very	Use -Pn to skip host discovery
slow Permission	Use -T4 or narrow target list
denied Incomplete	Use sudo or run as Admin
output	Add -v or --reason flags

Conclusion

Nmap is a must-know tool for any cybersecurity, network, or IT professional. With flexible scanning modes, firewall evasion techniques, and scripting capabilities, it allows deep insights into systems and services.

“Mastering Nmap = Mastering Network Visibility.”

Appendices

Common Ports

22 – SSH

23 – Telnet

25 – SMTP

53 – DNS

80 – HTTP

443 – HTTPS

3306 – MySQL

3389 – RDP

Nmap Cheatsheet

```
nmap -sS -T4 -p- -v 192.168.1.1
```

```
nmap -sV -sC 192.168.1.1
```

```
nmap --script=http-enum -p80 192.168.1.1
```